



Ratio and proportion

RATIO COMPARES THE SIZE OF QUANTITIES. PROPORTION COMPARES THE RELATIONSHIP BETWEEN TWO SETS OF QUANTITIES.

SEE ALSO

◀ 18–21 Multiplication

◀ 22–25 Division

◀ 48–55 Fractions

Ratios show how much bigger one thing is than another. Two things are in proportion when a change in one causes a related change in the other.

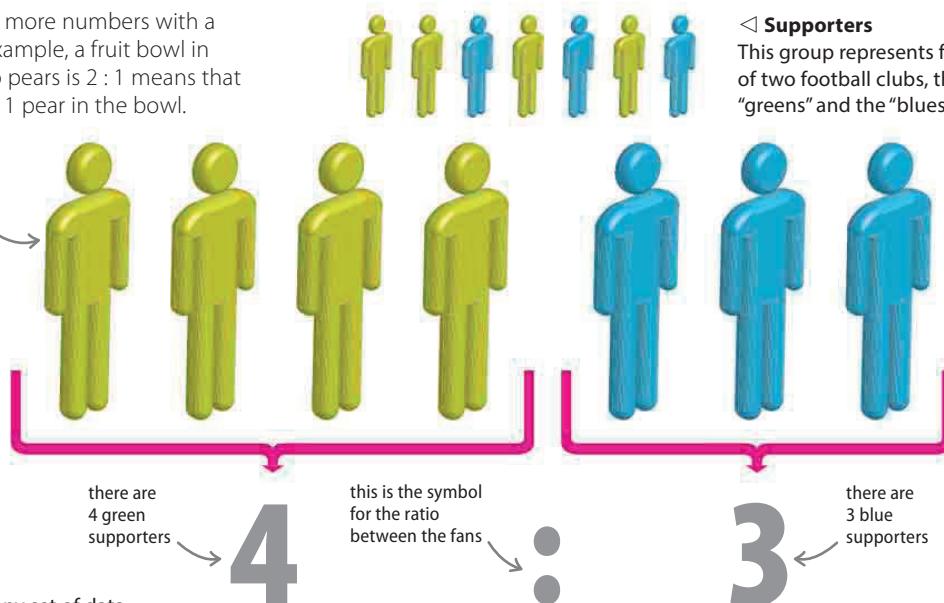
Writing ratios

Ratios are written as two or more numbers with a colon between each. For example, a fruit bowl in which the ratio of apples to pears is 2 : 1 means that there are 2 apples for every 1 pear in the bowl.

▷ Forming a ratio

To compare the numbers of people who support the two different clubs, write them as a ratio. This makes it clear that for every 4 green fans there are 3 blue fans.

these are the fans of the "greens"

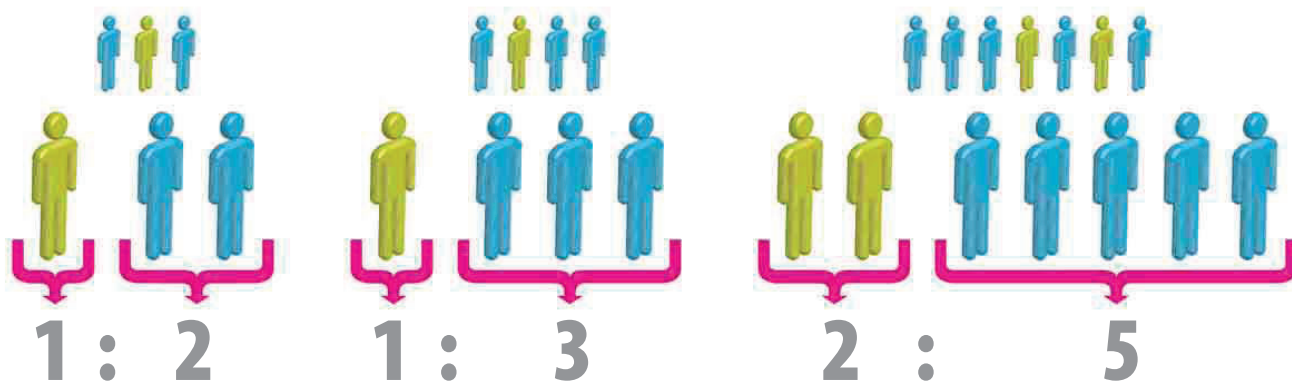


◁ Supporters

This group represents fans of two football clubs, the "greens" and the "blues."

▽ More ratios

The same process applies to any set of data that needs to be compared. Here are more groups of fans, and the ratios they represent.



△ 1 : 2

One fan of the greens and 2 fans of the blues can be compared as the ratio 1 : 2. This means that in this case there are twice as many fans of the blues as of the greens.

△ 1 : 3

One fan of the greens and 3 fans of the blues can be shown as the ratio 1 : 3, which means that, in this case, there are three times more blue fans than green fans.

△ 2 : 5

Two fans of the greens and 5 fans of the blues can be compared as the ratio 2 : 5. There are more than twice as many fans of the blues as of the greens.